

TALENTBRIDGE

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

<https://github.com/gankaiken/UM-Hackathon>

Table of Contents

1. Project Overview	1
2. Background & Business Objective	1
3. Product Purpose	2
4. System Functionalities	2
5. User Stories & Use Cases	3
6. Features Included (Scope Definition)	3
7. Features Not Included (Scope Control)	3
8. Assumptions & Constraints	4
9. Risks & Questions Throughout Development	4

Project Name: *TalentBridge AI*

Version: 1.0.1

1. Project Overview

- **Problem Statement:** Conventional recruitment is broken by two systemic failures:
 - *Credential Bias* (where talented individuals are filtered out by resume keywords or lack of "prestige" degrees) and ...
 - *Candidate Ghosting* (where the volume of applicants leads to zero feedback, damaging employer reputation).
- **Target Domain:** Human Resources Technology (HR-Tech), Recruitment, and Talent Acquisition.
- **Proposed Solution Summary:** *TalentBridge AI* is a Multi-Agent Recruitment Pipeline that replaces the resume screen with a structured, bias-blind AI interview. It uses a network of seven specialized AI agents (Mapper, Inquisitor, Strategist, etc.) to conduct and audit deep conversational assessments, producing a "Verdict Card" that scores candidates on real competency rather than paper credentials.

2. Background & Business Objective

- **Background of the Problem:** *TalentBridge AI* is a Multi-Agent Recruitment Pipeline that replaces the resume screen with a structured, bias-blind AI interview. It uses a network of seven specialized AI agents (Mapper, Inquisitor, Strategist, etc.) to conduct and audit deep conversational assessments, producing a "*Verdict Card*" that scores candidates on real competency rather than paper credentials.
- **Importance of Solving This Issue:**
 - Efficiency:

- HR teams currently spend 60% of their time screening resumes; *TalentBridge* automates this while providing 10x more depth than a standard form (The New York Times, n.d.).
- Strategic Fit / Impact:
 - In the age of AI, candidates are using LLMs to write perfect resumes. *TalentBridge* fights "AI with AI" by using a real-time behavioral monitor (The Sentinel) to ensure the integrity of the conversation, allowing companies to trust the data they receive.

3. Product Purpose

3.1. Main Goal of the System:

To provide a "Blind Audition" for the corporate world. By converting any job description into a dimension-based interview, the system extracts high-fidelity competency signals from candidates through a dignified, conversational experience.

3.2. Intended Users (Target Audience):

- Product Managers & Lead Designers:
 - Who needs to hire specialists but don't have time to interview every "decent-looking" resume.
- Founders of Early-Stage Startups:
 - Who need "T-shaped" talent (having knowledge in different sectors with specializing in one aspect) and want to verify problem-solving skills before the first human interview.
- Entrepreneurs & Small Business Owners:
 - Who lack internal HR departments and need a professional, automated way to screen for specific roles (e.g., Marketing, Ops).
- Frontend Developers & Low Code Builders:
 - Hiring for technical teams where "GitHub links" don't always tell the full story of collaborative ability.

4. System Functionalities

4.1. Description:

TalentBridge is a full-stack platform where employers post Job Descriptions (JD) and receive "Verdict Cards." Candidates apply by engaging in a text-based AI interview that adapts in real-time to their answers.

4.2. Key Functionalities:

- **AI Mapping:**
 - Instantly converts unstructured JDs into 5 core competency dimensions.
- **Adaptive Interviewing:**
 - A "Strategist" agent monitors the conversation and directs an "Inquisitor" to probe gaps or unexplored skills.
- **Sentinel Integrity Layer:**
 - Passionate browser-level tracking of tab switches, pastes, and focus loss to detect AI proxy usage.
- **Bias-Stript Auditing:**
 - A final "Auditor" agent reviews the full transcript, ignores grammar/slang, and scores the candidate purely on "Action Nodes" (logic, results, initiative).
- **Dignity-First Feedback:**
 - Candidates receive growth paths (upskilling) if they aren't a current match, ensuring a positive brand experience for the employer.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

Z.AI GLM-4. Chosen for its high linguistic flexibility (English, BM, Manglish) and strong instruction-following for structured JSON output.

4.3.2. Prompting Strategy:

Role-Specific Personas: Every agent has a narrow Scope of Work (SOW). The Auditor is told to be "the invisible final judge," while the Inquisitor is told to be "warm and adaptive."

Chain-of-Command: The Strategist outputs "Reasoning" blocks which the Inquisitor uses to formulate natural dialogue.

4.3.3. Context & Input Handling:

The system maintains a rolling window of the conversation, but injects "Sentinel Signals" as metadata. This allows the AI to perform a "Reality Check" (Sentinel Stage 2) if it detects the candidate is switching tabs to consult ChatGPT.

4.3.4. Fallback & Failure Behavior:

Schema Validator: A secondary agent validates the AI's scoring output. If the JSON is malformed or missing a score, it triggers a retry loop.

Mock Fallback: If the API key is missing or quotas are hit, the system gracefully falls back to structured mock data to ensure the UI remains functional for demos.

5. User Stories & Use Cases

- **User Stories:** *"As a Malaysian SME owner hiring a social media exec, I want to see if a candidate actually understands TikTok shop conversion logic, rather than just seeing 'worked at a shop' on their resume."*
- **Use Cases (Main Interactions):**
 - **The Post Flow:** Employer pastes a JD → Mapper creates dimensions → Dimension QA verifies quality → Unique interview link generated.

- **The Interview Flow:** Candidate enters name → Inquisitor starts chat → Sentinel watches behavior → Strategist ensures coverage
- **.The Audit Flow:** Session ends → Auditor strips bias → Verdict Card issued to HR Dashboard.
- **Use Case Diagram:** (TODO)

6. Features Included (Scope Definition)

- Full Employer Dashboard with KPI tracking (Time Saved, Reputation Score).
- Multi-agent AI backend (7 specialized agents).
- Real-time chat interface for candidates.
- Browser-level integrity monitoring (Sentinel).
- Drizzle ORM & SQLite database for session persistence.
- Stripe-inspired, premium "Enterprise" UI design.

7. Features Not Included (Scope Control)

- Voice/Video: Currently text-only to ensure faster processing and lower bias (accent/appearance).
- Resume Parsing: Consciously omitted; we want the JD to drive the interview, not a filtered CV.
- Third-Party ATS Sync: (Planned for v2).

8. Assumptions & Constraints

- Connectivity:
 - Requires stable internet for [Z.AI](#) streaming.
- Honesty:
 - Assumes that while candidates might use AI assistance, the Sentinel + Reality Check mechanism will catch 90% of automated proxy use.

9. Risks & Questions Throughout Development

- **Design Similarity:**

- **Risk:** Will common design tokens look generic? Solution: We implemented custom CSS variables and multi-layer shadows to achieve a bespoke "Finance-Tech" aesthetic.

- **Frame Stability:**

- **Risk:** Does the interview break on mobile? Solution: Used CSS ResizeObserver and Tailwind **responsive prefixes to ensure the chat window is usable on all screen sizes.**

- **Drawing Interpretation:**

- (N/A – We focus on text-based behavioural analysis).

- **AI Hallucination:**

- **Risk:** What if the Auditor makes up a score? Solution: The Schema Validator forces the Auditor to provide "Key Evidence" (a specific quote from the transcript) for every single score awarded. No quote = No score.

References

<https://www.nytimes.com/2025/10/07/business/ai-chatbot-prompts-resumes.html>